Distance between a line and point

Basic skills

Q1. a) $A = (0, 6)$ $B = (-3, 10)$

so vector between is $\underline{b} - \underline{a}$ ← or $\underline{a} - \underline{b}$

$$\underline{b} - \underline{a} = \begin{pmatrix} -3 \\ 10 \end{pmatrix} - \begin{pmatrix} 0 \\ 6 \end{pmatrix} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$$

← Pythagoras

so distance $\sqrt{(-3)^2 + 4^2} = 5$

b) $A = (5, 6, 2)$ $B = (-1, -2, 3)$

$$\underline{b} - \underline{a} = \begin{pmatrix} -6 \\ -8 \\ 1 \end{pmatrix} \text{ so } |\underline{b} - \underline{a}| = \sqrt{6^2 + 8^2 + 1^2} = \sqrt{101}$$

Q2. $\underline{a} = \begin{pmatrix} -2 \\ 5 \\ 0 \end{pmatrix}$ $\underline{b} = \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix}$

a) $$|\underline{b} - \underline{a}| = \left| \begin{matrix} 5 \\ 0 \\ 1 \end{matrix} \right| = \sqrt{5^2 + 0^2 + 1^2} = \sqrt{26}$$

b) $|\underline{b} - \underline{a}|$ is distance between A, B so
$|\underline{a} - \underline{b}|$ is distance between B, A so same

c) $\underline{a} \cdot \underline{b} = -2 \times 3 + 5 \times 5 + 0 \times 1 = 19$

now $|\underline{a}| = \sqrt{29}$, $|\underline{b}| = \sqrt{35}$

so $\cos\theta = \dfrac{19}{\sqrt{29}\sqrt{35}}$ ← $\cos\theta = \dfrac{\underline{a} \cdot \underline{b}}{|\underline{a}||\underline{b}|}$

$\theta = 53.4^{\circ}$

d) $(\widehat{\underline{b} - \underline{a}}) = \dfrac{\underline{b} - \underline{a}}{|\underline{b} - \underline{a}|}$

unit vector has magnitude 1.

$\hat{\underline{a}} = \dfrac{\underline{a}}{|\underline{a}|}$

$= \dfrac{1}{\sqrt{26}} \begin{pmatrix} 5 \\ 0 \\ 1 \end{pmatrix}$

e) direction is $\underline{b} - \underline{a}$ ← or $(\widehat{\underline{b} - \underline{a}})$, ...

starting is $\underline{a}$ ← or $\underline{b}$, ...

so $\underline{r} = \begin{pmatrix} -2 \\ 5 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 0 \\ 1 \end{pmatrix}$

Competency skills

Q3. $\underline{r} = \begin{pmatrix}1\\5\\2\end{pmatrix} + \lambda\begin{pmatrix}2\\3\\1\end{pmatrix}$ $P = \begin{pmatrix}3\\4\\5\end{pmatrix}$

a) $\begin{pmatrix}3\\4\\5\end{pmatrix} = \begin{pmatrix}1+2\lambda\\5+3\lambda\\2+\lambda\end{pmatrix}\begin{matrix}①\\②\\③\end{matrix}$ so $\lambda = 1$ from ①

but $\lambda = -\frac{1}{3}$ from ②

so no solution

so P is not on line

b) $\underline{r} = \begin{pmatrix}1+2\lambda\\5+3\lambda\\2+\lambda\end{pmatrix}$ so $(\underline{r}-\underline{P})\cdot\underline{d} = \begin{pmatrix}2\lambda-2\\3\lambda+1\\\lambda-3\end{pmatrix}\cdot\begin{pmatrix}2\\3\\1\end{pmatrix}$

$= 2(2\lambda-2)+3(3\lambda+1)+\lambda-3$

$= 4\lambda-4+9\lambda+3+\lambda-3$

$= 14\lambda-4$

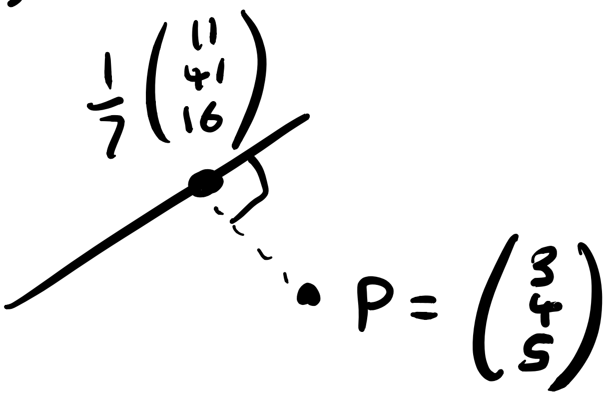
c) $\underline{a}\cdot\underline{b} = 0$ if perpendicular

solving $(\underline{r}-\underline{P})\cdot\underline{d} = 0$ means $14\lambda-4=0$

so $\lambda = \frac{4}{14} = \frac{2}{7}$

so point is $\underline{r} = \begin{pmatrix}1+2\times\frac{2}{7}\\5+3\times\frac{2}{7}\\2+\frac{2}{7}\end{pmatrix} = \frac{1}{7}\begin{pmatrix}11\\41\\16\end{pmatrix}$

d) Hence distance is



$$|\underline{r} - \underline{p}| = \left|\begin{pmatrix}-\frac{10}{7}\\ \frac{13}{7}\\ -\frac{19}{7}\end{pmatrix}\right|$$

$$= \frac{1}{7}\left|\begin{pmatrix}-10\\13\\-19\end{pmatrix}\right|$$

$$= \frac{1}{7}\sqrt{10^2+13^2+19^2} = \frac{3}{7}\sqrt{70} \approx 3.59$$

Q4. $\underline{r} = \begin{pmatrix}1-3\lambda\\ \lambda\\ 2-\lambda\end{pmatrix}$ $\quad \underline{p} = \begin{pmatrix}2\\5\\1\end{pmatrix}$

line as single vector

$$(\underline{r} - \underline{p}) \cdot \underline{d} = 0 \Rightarrow \begin{pmatrix}-3\lambda-1\\ \lambda-5\\ 1-\lambda\end{pmatrix} \cdot \begin{pmatrix}-3\\1\\-1\end{pmatrix} = 0$$

$$\Rightarrow 9\lambda + 3 + \lambda - 5 - 1 + \lambda = 0 \Rightarrow 11\lambda = 3$$

so $\lambda = \frac{3}{11}$ and $\underline{r} - \underline{p} = \begin{pmatrix}-3\times\frac{3}{11} - 1\\ \frac{3}{11} - 5\\ 1 - \frac{3}{11}\end{pmatrix} = \frac{1}{11}\begin{pmatrix}-20\\-52\\8\end{pmatrix}$

so $|\underline{r} - \underline{p}| = \frac{1}{11}\sqrt{20^2+52^2+8^2} = \frac{12\sqrt{22}}{11} \approx 5.12$

Q5. $\dfrac{x-2}{3} = \dfrac{y}{4} = \dfrac{z+3}{-1} = \lambda$

$x = 3\lambda + 2$
$y = 4\lambda$
$z = -3 - \lambda$

in vector form $\underline{r} = \begin{pmatrix} 2 \\ 0 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 4 \\ -1 \end{pmatrix} = \begin{pmatrix} 2+3\lambda \\ 4\lambda \\ -3-\lambda \end{pmatrix}$

$\underline{p} = \begin{pmatrix} 3 \\ -10 \\ 1 \end{pmatrix}$

$(\underline{r} - \underline{p}) \cdot \underline{d} = 0 \Rightarrow \begin{pmatrix} 3\lambda - 1 \\ 4\lambda + 10 \\ -4 - \lambda \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 4 \\ -1 \end{pmatrix} = 0$

$\Rightarrow 9\lambda - 3 + 16\lambda + 40 + 4 + \lambda = 0 \Rightarrow 26\lambda = -41$

so $\lambda = -\dfrac{41}{26}$ so $\underline{r} = \begin{pmatrix} 2 + 3 \times -\frac{41}{26} \\ 4 \times -\frac{41}{26} \\ -3 - -\frac{41}{26} \end{pmatrix} = \dfrac{1}{26}\begin{pmatrix} -71 \\ -164 \\ -37 \end{pmatrix}$

Advanced skills

Q6. Line is $x = y = z$ so

$$\underline{r} = \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \lambda \\ \lambda \\ \lambda \end{pmatrix}$$

any coordinate is $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$

so $(\underline{r} - \underline{p}) \cdot \underline{d} = 0 \Rightarrow \begin{pmatrix} \lambda - x \\ \lambda - y \\ \lambda - z \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 0$

$$\Rightarrow \quad \lambda - x + \lambda - y + \lambda - z = 0$$

$$\Rightarrow \quad 3\lambda = x + y + z \quad \Rightarrow \quad \lambda = \frac{x+y+z}{3}$$

so

$$\underline{r} - \underline{p} = \frac{1}{3}\begin{pmatrix} -2x + y + z \\ x - 2y + z \\ x + y - 2z \end{pmatrix}$$

so distance is

$$\frac{1}{3}\sqrt{(-2x+y+z)^2 + (x-2y+z)^2 + (x+y-2z)^2}$$

notice on $x=y=z$ distance is zero

Q7.

$$\underline{r} = \begin{pmatrix} 1+\lambda \\ 1-2\lambda \\ 4-5\lambda \end{pmatrix} \qquad \underline{a} = \begin{pmatrix} \alpha \\ -1 \\ 1 \end{pmatrix}$$

$$(\underline{r} - \underline{a}) \cdot \underline{d} = 0 \Rightarrow \begin{pmatrix} 1+\lambda-\alpha \\ 2-2\lambda \\ 3-5\lambda \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ -5 \end{pmatrix} = 0$$

$$\Rightarrow \quad 1 + \lambda - \alpha - 4 + 4\lambda - 15 + 25\lambda = 0$$

$$\text{so} \quad 30\lambda = \alpha + 18 \qquad \lambda = \frac{\alpha + 18}{30}$$

Then

$$\underline{r}-\underline{a}=\frac{1}{30}\begin{pmatrix}48-29\alpha\\24-2\alpha\\-5\alpha\end{pmatrix}$$

so distance is $\frac{1}{30}\sqrt{(48-29\alpha)^2+(24-2\alpha)^2+(-5\alpha)^2}$

minimal when $(48-29\alpha)^2+(24-2\alpha)^2+(-5\alpha)^2$ is minimal. So Differentiate

$$2\times-29(48-29\alpha)+2\times-2(24-2\alpha)+2\times-5\times-5\alpha$$

$$-2784+1682\alpha-96+8\alpha+50\alpha$$

for min need stationary so $1740\alpha=2880$

$$\Rightarrow \alpha=\frac{48}{29}$$

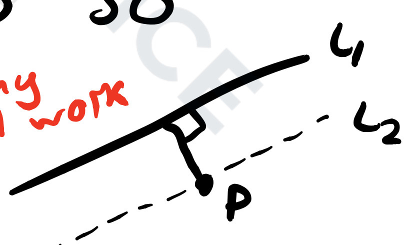
for quadratic stationary is min

Q8. $\underline{r}_1=\begin{pmatrix}2+3\lambda\\2+\lambda\\1-2\lambda\end{pmatrix}$ $\underline{r}_2=\begin{pmatrix}1-9\mu\\-3\mu\\1+6\mu\end{pmatrix}$

Let point P be $\mu=0$ so

$$\underline{P}=\begin{pmatrix}1\\0\\1\end{pmatrix}$$

any will work



from line 2

$(\underline{r}_1-\underline{P})\cdot\underline{d}_1=0$ $\Rightarrow \begin{pmatrix}1+3\lambda\\2+\lambda\\-2\lambda\end{pmatrix}\cdot\begin{pmatrix}3\\1\\-2\end{pmatrix}=0$

$\Rightarrow 3+9\lambda+2+\lambda+4\lambda=0 \Rightarrow 14\lambda=-5$

$\Rightarrow \lambda = -\frac{5}{14}$ so $\underline{r}_1 - \underline{p} = \frac{1}{14}\begin{pmatrix} -1 \\ 23 \\ 10 \end{pmatrix}$

distance is therefore

$$\frac{1}{14}\sqrt{1^2+23^2+10^2} = \frac{3\sqrt{70}}{14} \approx 1.793$$

check another point if you want to check answer